

ARAVIND KUMAR MADAVARAM

DATA SCIENTIST | PyTorch • Deep Learning • Computer Vision • MLOps

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PROFESSIONAL SUMMARY

Data Scientist building deep learning pipelines for biomedical computer vision — 2D and 3D instance segmentation, quantitative image analysis, and end-to-end model training and inference in PyTorch on NVIDIA GPU. Prior industry background in large-scale data analytics at Amazon Payments and ICICI Bank, spanning fraud/risk investigations and enterprise reporting. Comfortable owning the workflow end-to-end from raw data through model training, evaluation, and downstream analytics — contributed to one peer-reviewed publication.

TECHNICAL SKILLS

Deep Learning & Computer Vision: PyTorch, Cellpose, Omnipose, Cellpose-SAM, BiaPy, CNN architectures, 2D/3D instance segmentation, semantic segmentation, transfer learning, model training & evaluation, GPU inference

Python & ML Stack: Python, NumPy, Pandas, scikit-learn, PySpark, scikit-image, tifffile, image analytics, quantitative analysis

MLOps & Infrastructure: Git, Docker, conda environments, NVIDIA CUDA GPU pipelines, pytest, reproducible training/inference workflows, CI-friendly project structure

Data & Analytics: SQL Server, MySQL, Hive/Hadoop, SSIS/ETL, Tableau, Power BI, root-cause analysis, fraud/risk pattern detection

PROFESSIONAL EXPERIENCE

Data Analyst | UofL Health, Department of Ophthalmology

Louisville, KY | May 2024 – Present

- **Deep Learning (PyTorch, CNN architectures):** Trained and evaluated 3D instance segmentation models on high-resolution biomedical imaging volumes across multiple experimental datasets, producing per-instance masks used for downstream quantitative analysis.
- **Computer Vision (Cellpose, Omnipose, Cellpose-SAM, BiaPy):** Built and benchmarked 2D and 3D segmentation pipelines on high-resolution microscopy volumes; configured custom conda / Python 3.11 / CUDA environments, added auto-crop for background regions at inference, and ran comparative evaluation across model architectures.
- **3D Image Analysis (Python, scikit-image):** Developed a post-segmentation quantitative analysis pipeline computing local thickness maps, skeleton-based structural detection, surface mesh measurements, and per-instance CSV outputs for downstream research use.
- **MLOps & Reproducibility (Git, Docker, conda, pytest):** Standardized training and inference workflows for reproducibility across concurrent projects; open-sourced an in-house biomedical image conversion pipeline (github.com/UofL-McCall-lab/oct-ome-metadata-converter) replacing paid third-party software, backed by a pytest QC suite.
- **Data Engineering (SQL Server, Python, SSIS):** Designed schemas, CTE queries, stored procedures, and Python / SSIS ETL pipelines connecting clinical and research systems; automated recurring reporting, improving repeatable processing efficiency by 40%. Contributions supported one peer-reviewed publication.

Transaction Risk Analyst | Amazon, Payments & E-commerce

Hyderabad, India | Jul 2021 – Jan 2023

- **Large-Scale Data (SQL, Hive/Hadoop, MySQL):** Queried and joined millions of customer, transaction, and behavioral records across payments infrastructure to surface fraud patterns, risk drivers, and account-takeover signals.
- **Analytics & Reporting (Tableau, Power BI):** Built KPI scorecards and recurring dashboards for operations leadership tracking investigation volumes, handling efficiency, defect patterns, and account-integrity trends.
- **Root-Cause Analysis:** Led investigations on high-risk transaction workflows; process changes from these analyses drove a 4.8% operational efficiency improvement.

Data Analyst | ICICI Bank, Payments & Financial Services

Mumbai, India | Nov 2020 – Jul 2021

- **SQL & Advanced Excel:** Analyzed customer, account, payment, and service data across four financial-services domains, building KPI, trend, and exception reports for operational decision-making.
- **Power BI & Data Profiling:** Built visualizations and ran reconciliation checks across enterprise systems, improving accuracy of downstream leadership reports and surfacing compliance indicators to non-technical stakeholders.

Business Analyst | Assert Tech Solutions

Hyderabad, India | Jul 2017 – Jun 2020

- **SQL & Requirements Analysis:** Translated stakeholder needs into business requirements, workflow maps, and SQL reporting specifications for technical and non-technical teams across multiple client engagements.
- **Python & Excel Automation:** Automated recurring SQL, Excel, and Python reporting workflows, saving 5+ hours per week while improving delivery consistency.

EDUCATION

MS Business Analytics (Machine Learning), University of Louisville | 2024

MBA, Operations Management, Institute of Public Enterprise, Hyderabad | 2018 – 2020

BTech Electronics & Communication Engineering, VIT | 2013 – 2017

CERTIFICATIONS & PUBLICATION

Certifications: Machine Learning Specialization (DeepLearning.AI / Stanford Online) • Google Analytics

Publication: Co-author, peer-reviewed ophthalmology research publication, 2024